

HR Workforce Analytics Using Python & Tableau

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1. Executive Summary

This project simulates a real-world HR analytics system by generating synthetic employee data using Python and analyzing workforce patterns using Tableau dashboards. The analysis focuses on workforce demographics, hiring trends, compensation patterns, performance distribution, and employee attrition.

The objective is to demonstrate people analytics capabilities and provide insights that support HR strategy, workforce planning, and organizational decision-making.

2. Business Objectives

The project was conducted to:

- Analyze workforce demographics and diversity
 - Monitor hiring and termination trends
 - Understand salary and compensation patterns
 - Evaluate performance distribution across education levels
 - Identify geographic workforce distribution
 - Provide insights for HR strategy and workforce optimization
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3. Data Generation & Methodology

Synthetic Dataset Generation (Python)

A synthetic HR dataset was generated using Python with realistic business rules to simulate corporate workforce data.

Key Simulation Logic

- Probabilistic department and job role assignment
- Education constraints per job role
- Salary ranges per job title with education and gender modifiers

- Hiring year distribution to simulate workforce growth
- Termination simulation (~11% attrition)
- Performance ratings distribution
- Demographic distribution modeling (gender, age groups, education levels)

This approach enables realistic people analytics without exposing real employee data.

4. Data Model & Architecture

The dataset follows a structured analytical model:

Fact Table

- Employee-level records containing salary, performance, tenure, termination status, and demographics.

Dimension Attributes

- Gender
- Age and Age Groups
- Education Level
- Department and Job Title
- Location (State)
- Hire and Termination Dates

This structure supports multi-dimensional HR analysis across demographics, time, and geography.

5. Key Workforce Metrics

Workforce Overview

- Total Employees: **8,950**
- Active Employees: **~7,950**
- Terminated Employees: **~1,000**

Demographics

- Gender Distribution: Balanced male–female workforce
- Age Distribution: Majority in mid-career age groups

- Education Levels: Bachelor's and Master's degrees dominate

Hiring & Attrition

- Hiring trend shows steady workforce growth over years
- Terminations remain stable with slight year-over-year variation

Compensation

- Salary increases with age and job seniority
 - Higher education correlates with higher salary bands
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6. Workforce Demographics & Diversity Insights

Gender & Education

- Gender distribution is relatively balanced across education levels
- Higher education levels are more prevalent in technical and managerial roles

Age & Education

- Younger employees primarily hold Bachelor's degrees
- Senior employees show higher Master's and Doctorate representation

Implication: Workforce education level increases with tenure and experience.

7. Organizational Structure & Hiring Trends

Department Distribution

- Core departments include IT, HR, Finance, Sales, and Operations
- IT and Sales employ the largest workforce share

Hiring Trends

- Hiring increased significantly in recent years, indicating organizational expansion
 - Recent hiring peaks suggest business growth or digital transformation initiatives
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8. Compensation & Performance Analysis

Age vs Salary

- Salary shows a positive correlation with age and experience
- Senior employees earn significantly higher compensation

Education vs Performance

- Higher education levels correlate with improved performance ratings
- Employees with Master's and Doctorate degrees show higher average performance scores

Implication: Investment in education and training can improve workforce productivity.

9. Employee Attrition Analysis

Termination Trends

- Terminations remain relatively stable across years
- Attrition rate is moderate (~11%), typical for large organizations

Geographic Distribution

- Workforce is concentrated in major US states
- Certain states show higher employee density, indicating regional hubs

Implication: Geographic workforce concentration can impact regional HR policies and office infrastructure planning.

10. Key Business Insights

- Workforce demographics are balanced across gender and age groups
 - Education level strongly influences salary and performance outcomes
 - Hiring growth indicates expanding organizational scale
 - Attrition is stable but requires monitoring to prevent talent loss
 - Geographic workforce clustering suggests centralized operational regions
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11. Strategic HR Recommendations

Workforce Strategy

- Implement targeted retention programs for high-performing senior employees
- Use predictive attrition modeling to reduce turnover

Compensation Strategy

- Align salary structures with education and performance metrics
- Introduce performance-based incentives for productivity enhancement

Talent Development

- Encourage higher education and certifications for career progression
- Invest in training programs to improve performance outcomes

Geographic HR Planning

- Strengthen HR policies in high-density states
 - Consider regional talent acquisition strategies
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12. Tools & Skills Demonstrated

- Python (Synthetic Data Generation, Faker, NumPy, Pandas)
 - Tableau (HR Dashboards, Filters, Calculated Fields)
 - People Analytics & Workforce Analytics
 - Data Modeling & Simulation
 - Business Intelligence & Data Storytelling
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13. Deliverables

- Synthetic HR dataset (CSV)
 - Python data generation script
 - Tableau dashboards (HR Summary & Details)
 - Analytical PDF report
 - Project documentation and Tableau Public link
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Tableau Public Dashboard Link

https://public.tableau.com/views/HRDashboard_17695890972600/HRSummary?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link